

Enabling Advanced Interactions through Closed-loop Control of Motor Unit Activity After Tetraplegia

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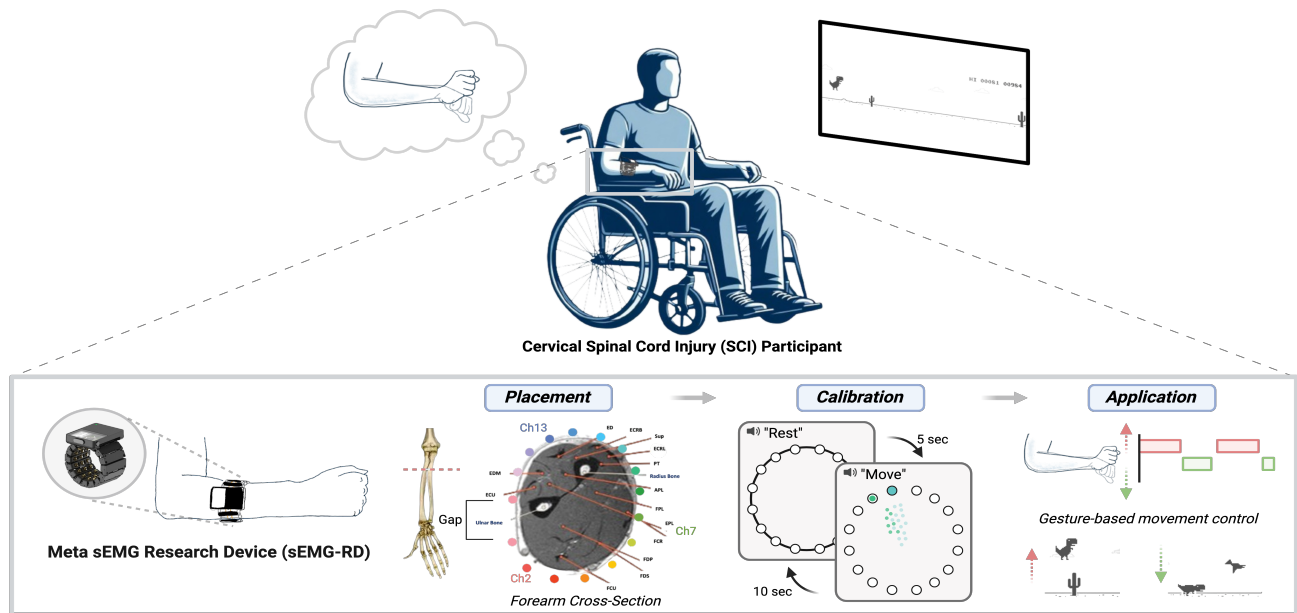


Figure 1: A cervical spinal cord injury (SCI) participant using a non-invasive surface electromyography (sEMG) interface, manufactured by Meta Reality Labs, to control a cursor / 2D character's movement in a computer system by utilizing residual muscle activity from attempted movement.

ABSTRACT

Studies have shown that most individuals with motor complete spinal cord injuries (SCIs) can activate motor units (MUs) below the injury level, potentially enabling the use of myoelectric controllers

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for computer tasks. We present a novel wearable neuromotor interface driven by surface electromyography (sEMG). This wearable sEMG interface detects motor unit action potentials (MUAPs) from an array of sensors on the forearm. The MU event rate is translated (i.e. decoded) into discrete or continuous inputs for emulating button presses or joystick inputs. Our study demonstrates real-time detection and decoding of MUAPs in two individuals with tetraplegia, enabling computer task control via a non-invasive interface. Participants calibrated a spike sorting model through periods of rest and tonic activation, mapping MUAP firing rates to various degrees of freedom (DOF) for cursor or character control in tasks and 2D games. Each MUAP was mapped to a separate DOF axis, which we

utilized in a Fitts' Law target acquisition task to evaluate device throughput rate and other performance metrics such as completion rate, initiation time, time to target acquisition, and dial-in time. Our findings highlight MU firing as an effective control input, enabling gaming and social interaction for individuals with tetraplegia. This device's wearability and ease of use offer an innovative human-computer interaction solution that may enable people with SCI to interact freely with computers and other digital devices.

KEYWORDS

Human-Computer Interaction (HCI); Surface Electromyography (sEMG); Cervical Spinal Cord Injury (SCI), Tetraplegia; Myoelectric Control; Wearable Neuromotor Interface; Motor Units; Motor Unit Firing; Firing Rate; Fitts' Law;

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1 INTRODUCTION

A growing number of studies have demonstrated that most people with motor complete spinal cord injuries (SCIs) retain the ability to activate motor units (MUs) below the level of injury, offering the potential to use myoelectric controllers for computer-based tasks [5] [7] [6] [2] [4]. Building upon this knowledge, we present a novel wearable neuromotor interface driven by surface electromyography (sEMG) [3]. The interface is packaged in a wristband form factor that is easy to don and doff and designed for seamless plug-and-play functionality. The wristband contains 16 differential input channels distributed circumferentially around the wristband for measuring sEMG and detecting motor unit action potentials (MUAPs). Unlike other EMG devices with bracelet-type 8 channel array electrode configurations (MYO band). The strength of a muscle contraction increases with the rate of MUAP events (i.e. firing rate) occurring in the muscle. Thus, detecting and discriminating intended actions from the pattern of MUAP firing measured in a group of muscles is possible. This way, the wristband can be an alternative interface modality for human-computer interaction (HCI) [1]. Such an interface is beneficial for people with motor impairments who are unable to generate sufficient muscle force to move their hand to type on a keyboard or use a mouse. In this study, we explored the potential of this HCI modality as an accessibility interface for people with severe motor impairment (see Fig. 1). Specifically, we tested the performance of two individuals with tetraplegia (AIS A and B) using the device to perform 1- or 2-degree of freedom (DOF) computer-based tasks.

2 INPUT CONTROL SELECTION AND CALIBRATION

Participants were seated in front of a computer display that provided visual instructions and prompts to perform various attempted

movements of the wrist and fingers. At the start of each session, a screening and calibration run was performed to isolate MUAPs and set parameters for a spike sorting model. In subsequent runs, the MUAPs were detected in real-time and the instantaneous firing rate (MUAP events/sec) was calculated and smoothed with an exponential moving average filter. The firing rates for individual MUs were mapped to different DOF for cursor/character control in 1D and 2D cursor-control tasks and games. While we could isolate more than two MUAPs per session, we required a minimum of two MUAPs for the experiments described here. Participants selected at least two gestures for calibrating the MUAP detection model. The calibration interface cycled through each intended gesture individually, prompting the user to alternate between actively attempting the intended gesture (without movement) and rest as instructed. An internal classification algorithm then selects the best candidate MU source for an intended gesture once the calibration is complete. Finally, a visual display with a rasterized indicator of the MU activation attempts allowed participants to volitionally modulate the detected MUAP, allowing the user to test ease of use and control quickly.

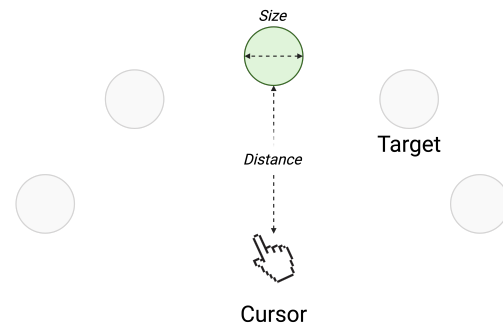


Figure 2: A Fitts' Law-style target acquisition task was selected as the baseline evaluation method.

3 EVALUATION OF FUNCTIONAL TASK THROUGHPUT AS ANALOG CONTROLLER

After calibration, each MUAP was mapped to a separate DOF axis according to the user's preference. Per ISO 9241 recommendation, we used a variant of a Fitts' Law task (Fitts 1954; Fig. 2) to characterize the device throughput rate. We required the subject to move the cursor from a fixed starting location to target regions of varying size and distance from the start position, enabling only one axis per target (non-selective control task) or both axes simultaneously (selective control task). Each MUAP control axis mapping was scaled such that the cursor position varied in linear proportion to the firing rate of the MUAP, with the maximum firing rate corresponding to the position limit (e.g. screen width) of the cursor. Participants were required to control the MUAP firing rate to move the cursor to one of three target locations and hold the cursor within the target boundary for at least 1 sec. Small, medium, and large targets were used to vary the accuracy constraint, and a trial was considered successful if participants captured the target within 15 sec. Performance metrics included **completion rate** (percentage of trials successfully finished), **initiation time** (time taken to start moving towards a target after trial start), **time to target acquisition** (time taken from initial movement to first contact with the target), and

dial-in time (time taken to settle on the target and complete the task). Both participants demonstrated improvements in completion rate, initiation time, time to target acquisition, and dial-in time over repeated sessions, which was particularly evident during the non-selective control task.

4 EVALUATION OF DYNAMIC TASK AS PUSHBUTTON CONTROLLER

In a few sessions, time was allotted for pilot interactions with recreational/dynamic tasks. The first dynamic task required only a single MUAP to control the vertical position of a character to avoid colliding with obstacles scrolling horizontally toward the character (i.e. a “Flappy-Bird” style game). A second game required control of two motor units and the firing rate of each was mapped nonlinearly to two gamepad buttons. Activating a button required increasing the firing rate for the associated motor unit above a preset threshold while preventing the other motor unit from exceeding its threshold. The player used the button inputs to control a spaceship along the horizontal axis using a stateful velocity encoding to update the position of the ship along the axis, requiring the player to navigate beneath oncoming targets of varying sizes and for varying durations (i.e. a “Space Shooter” style game). In each game, the High Score of all players as well as specified metrics based on obstacle and target performance was recorded for analysis.

In summary, these findings highlight the potential of MU firing as control inputs for computer tasks, enabling individuals with

tetraplegia to participate in gaming and social interactions. This device’s wearability and ease of use offer an innovative HCI solution. By leveraging residual MU and sEMG activity, this innovation promises to enhance the quality of life for those with SCI.

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